

HealthSync — Quality Assurance Test Report

Version: 1.0.0 **Date:** 19 July 2026 **Tester:** Oyekunle Oyekola **Environment:** macOS Darwin 25.4.0 / Node.js 22.x / SQLite (dev) / Next.js 16.2.10 **Classification:** Internal — QA Team

1. Executive Summary

A total of **62 test cases** were executed across 8 functional modules. **58 passed, 4 failed**, and all 4 failures have been **remediated** in this release. The application meets the acceptance criteria defined in the HealthSync Development Specification.

Status	Count	Percentage
PASS	58	93.5%
FAIL (remediated)	4	6.5%
BLOCKED	0	0%
Total	62	100%

2. Test Scope

Modules Tested

Module	Test Cases	Status
Authentication & Login	10	All Pass
Patient Registration	8	All Pass
Appointment Scheduling	9	All Pass
Clinical Encounters (SOAP)	8	All Pass
Ward & Bed Management	7	All Pass
Patient Portal	6	All Pass
RBAC & Permissions	8	All Pass
Audit Trail	6	All Pass

Out of Scope

- Billing module (Phase 2)
- Pharmacy module (Phase 2)
- Laboratory module (Phase 2)
- Reporting & Analytics (Phase 2)

3. Test Results by Module

3.1 Authentication & Login

ID	Test Case	Steps	Expected	Actual	Status
AUTH-01	Valid staff login	Enter admin@healthsync.io / Password123!	Redirect to /dashboard	Redirected to /dashboard	PASS
AUTH-02	Valid patient login	Enter portal@healthsync.io / Password123!	Redirect to /portal	Redirected to /portal	PASS
AUTH-03	Invalid email	Enter fake@example.com / password	"Invalid email or password"	"Invalid email or password"	PASS
AUTH-04	Invalid password	Enter admin@healthsync.io / wrongpass	"Invalid email or password"	"Invalid email or password"	PASS
AUTH-05	Identical error for invalid email and password	Compare AUTH-03 and AUTH-04 error messages	Same message (no user enumeration)	Same message	PASS
AUTH-06	Inactive user blocked	Deactivate user, attempt login	"Invalid email or password"	"Invalid email or password"	PASS
AUTH-07	Session persists across page loads	Login, navigate between pages	Session maintained	Session maintained	PASS
AUTH-08	Logout clears session	Click sign out	Redirect to /login, session cookie deleted	Redirected, cookie cleared	PASS
AUTH-09	Unauthenticated access blocked	Visit /dashboard without login	Redirect to /login	Redirected to /login	PASS
AUTH-10	Rate limiting on login	Submit 6 rapid failed logins	"Too many login attempts" message	"Too many login attempts" message	PASS

3.2 Patient Registration

ID	Test Case	Steps	Expected	Actual	Status
PAT-01	Register new patient	Fill all required fields	Patient created with MRN	Patient HS-2026-XXXXX created	PASS

PAT-02	MRN auto-generation	Register patient	MRN in HS-YYYY-XXXXX format	HS-2026-00005 generated	PASS
PAT-03	MRN uniqueness	Register 5 patients	All MRNs unique	All unique, sequential	PASS
PAT-04	Missing required fields	Submit with empty first name	Validation error	"First name is required"	PASS
PAT-05	Future date of birth rejected	Enter DOB = 2030-01-01	Validation error	"Date of birth must be a valid past date"	PASS
PAT-06	National ID masked	Enter full national ID	Only last 4 digits stored	Stored as last 4 only, displayed as ".... XXXX"	PASS
PAT-07	Allergy entry (multi-line)	Enter "Penicillin\nAspirin"	Two allergy entries created	Two entries in JSON array	PASS
PAT-08	Patient search by name/MRN/phone	Search various criteria	Matching results returned	Correct results for name, MRN, phone	PASS

3.3 Appointment Scheduling

ID	Test Case	Steps	Expected	Actual	Status
APT-01	Book appointment	Select patient, doctor, date, time	Appointment created	Created with status "booked"	PASS
APT-02	Slot availability shown	Select doctor and date	Available/taken slots displayed	Slots correctly marked	PASS
APT-03	No double-booking	Book same doctor, same time	"That slot is already booked"	Error returned, booking rejected	PASS
APT-04	Past appointment rejected	Select a past date/time	"Appointments cannot be booked in the past"	Validation error shown	PASS
APT-05	Cancel appointment	Cancel with reason	Status changed, reason logged	Status "cancelled", reason stored	PASS
APT-06	Cancel requires reason	Cancel without reason	Validation error	"A cancellation reason is required"	PASS

APT-07	Complete appointment	Mark as completed	Status changed to "completed"	Status updated	PASS
APT-08	Cancelled slot freed	Cancel, then book same slot	Booking succeeds	Slot available again	PASS
APT-09	Date navigation	Navigate to different dates	Correct appointments shown	Appointments filtered by date	PASS

3.4 Clinical Encounters (SOAP Notes)

ID	Test Case	Steps	Expected	Actual	Status
ENC-01	Create SOAP note	Fill S, O, A, P fields	Medical record created	Record with all SOAP fields saved	PASS
ENC-02	Add diagnoses	Enter "I10 Essential hypertension"	ICD-10 entry parsed	Parsed into structured JSON	PASS
ENC-03	Add prescriptions	Enter "Amoxicillin 500mg TDS 7 days"	Prescription entry created	Structured prescription saved	PASS
ENC-04	CDS allergy alert (client)	Prescribe amoxicillin to penicillin-allergic patient	Live allergy alert shown	Critical alert displayed immediately	PASS
ENC-05	CDS allergy alert (server)	Submit amoxicillin prescription via form	Server blocks without override	"Blocked by clinical decision support"	PASS
ENC-06	CDS override with acknowledgment	Check override box, submit	Record created, override logged in audit	Record saved, audit entry with overrodeAlerts:true	PASS
ENC-07	Drug interaction alert	Prescribe warfarin + aspirin	Critical interaction alert	Alert: "Markedly increased bleeding risk"	PASS
ENC-08	Sign medical record	Check "Sign this note"	signedAt and signedBy populated	Digital signature recorded	PASS

3.5 Ward & Bed Management

ID	Test Case	Steps	Expected	Actual	Status
WRD-01	View bed board	Navigate to /wards	Beds shown by ward with status	Color-coded bed grid displayed	PASS
WRD-02	Admit patient	Select patient and available bed	Admission created, bed marked occupied	Transaction: admission + bed + patient status updated	PASS
WRD-03	Reject admission to occupied bed	Admit to already-occupied bed	"That bed is no longer available"	Error returned	PASS
WRD-04	Reject duplicate admission	Admit already-admitted patient	"Patient already has an active admission"	Error returned	PASS
WRD-05	Discharge patient	Click discharge	Bed freed, patient status updated	Transaction: admission discharged, bed available, patient discharged	PASS
WRD-06	Bed status colors	View bed board	Available=green, Occupied=blue, Maintenance=gray	Correct color coding	PASS
WRD-07	Current inpatients list	View /wards	Active admissions listed	All active admissions shown with bed/ward info	PASS

3.6 Patient Portal

ID	Test Case	Steps	Expected	Actual	Status
PRT-01	Portal login	Login as patient	Redirect to /portal	Redirected to /portal	PASS
PRT-02	Staff cannot access portal	Login as admin, navigate to /portal	Redirect to /dashboard	Redirected away	PASS
PRT-03	See own appointments only	View portal	Only own appointments shown	Scoped to session.patientId	PASS
PRT-04	See signed records only	View medical records	Draft records hidden	Only records with signedAt shown	PASS
PRT-05	See latest vitals	View observations	Most recent vitals displayed	Latest vitals shown	PASS

		section			
PRT-06	See prescribed medications	View medical records	Prescriptions listed	Drug, dose, frequency, duration shown	PASS

3.7 RBAC & Permissions

ID	Test Case	Steps	Expected	Actual	Status
RBAC-01	Admin sees all nav items	Login as admin	Dashboard, Patients, Appointments, Wards, Staff, Audit	All 6 items visible	PASS
RBAC-02	Nurse cannot prescribe	Login as nurse, create encounter with prescription	403 error	"Role nurse is not permitted to prescribe"	PASS
RBAC-03	Nurse cannot see staff page	Login as nurse, navigate to /staff	Redirect or 403	Staff link not shown; direct URL returns 403	PASS
RBAC-04	Billing sees limited nav	Login as billing	Dashboard, Patients, Appointments only	Only permitted items shown	PASS
RBAC-05	Audit log admin-only	Login as doctor, navigate to /audit	Access denied	403 returned	PASS
RBAC-06	Doctor scope on dashboard	Login as doctor	"My appointments today"	Own appointments only, not hospital-wide	PASS
RBAC-07	Server-side RBAC enforcement	Craft direct POST to protected action	403 response	AuthError(403) thrown	PASS
RBAC-08	Permission matrix display	View /staff page	Visual permission matrix	All 13 permissions x 6 roles displayed	PASS

3.8 Audit Trail

ID	Test Case	Steps	Expected	Actual	Status
AUD-01	Login logged	Login as any user	auth.login entry created	Entry with actor, IP, timestamp	PASS

AUD-02	Patient creation logged	Register patient	patient.create entry	Entry with MRN in detail	PASS
AUD-03	Appointment logged	Book appointment	appointment.create entry	Entry with patient/doctor IDs	PASS
AUD-04	Record signing logged	Sign a medical record	record.sign entry	Entry with record ID	PASS
AUD-05	CDS override logged	Override critical alert	record.create with overrodeAlerts:true	Override documented in detail JSON	PASS
AUD-06	Audit log filterable	Filter by action/user	Filtered results shown	Correct filtering	PASS

4. Failed Test Cases and Remediation

FAIL-01: No Rate Limiting on Login (AUTH-10)

- **Original Result:** No protection against brute-force; unlimited login attempts accepted
- **Severity:** High
- **Remediation:** Implemented `src/lib/rate-limit.ts` with 5 attempts per 15-minute window, 30-minute lockout. Rate limit events logged to audit trail.
- **Retest:** PASS — 6th attempt returns "Too many login attempts" message

FAIL-02: Missing Security Headers

- **Original Result:** No X-Content-Type-Options, X-Frame-Options, CSP, or HSTS headers
- **Severity:** High
- **Remediation:** Added `src/middleware.ts` with OWASP-recommended security headers including CSP, X-Frame-Options: DENY, Referrer-Policy, and HSTS (production only).
- **Retest:** PASS — All headers present in response

FAIL-03: Weak JWT Secret Accepted in Production

- **Original Result:** Application accepted short/predictable JWT secrets without warning
- **Severity:** Critical
- **Remediation:** Added validation in `src/lib/auth.ts` — production mode requires min 32 characters and rejects strings containing "dev-secret".
- **Retest:** PASS — Application refuses to start with weak secret in production mode

FAIL-04: No Input Sanitization on Patient Text Fields

- **Original Result:** HTML tags and `javascript:` URIs stored verbatim in patient text fields
- **Severity:** Medium
- **Remediation:** Added `src/lib/sanitize.ts` with `sanitizeFormData()` applied to patient registration. Strips `<>`, `javascript:`, and `on*` event handlers.
- **Retest:** PASS — HTML tags stripped from stored data

5. Automated Acceptance Tests

The `scripts/verify.ts` suite runs 21 programmatic checks. All pass:

```
[PASS] 1. MRN format matches HS-YYYY-XXXXX
[PASS] 2. All MRNs are unique
[PASS] 3. No double-booked appointments exist
[PASS] 4. Cancelled appointments have reasons
[PASS] 5. Admin has staff:manage permission
[PASS] 6. Doctor has prescribe permission
[PASS] 7. Nurse lacks prescribe permission
[PASS] 8. Patient has no staff permissions
[PASS] 9. Billing lacks record:write
[PASS] 10. Frontdesk has appointment:write
[PASS] 11. Only admin has audit:read
[PASS] 12. CDS detects penicillin cross-reactivity
[PASS] 13. CDS detects warfarin+aspirin interaction
[PASS] 14. CDS detects simvastatin+clarithromycin
[PASS] 15. Audit log has entries for all mutation types
[PASS] 16. All occupied beds have active admissions
[PASS] 17. All active admissions reference occupied beds
[PASS] 18. Signed records have signedBy populated
[PASS] 19. All patients have valid status values
[PASS] 20. All beds have valid status values
[PASS] 21. All appointments have valid status values
```

6. Non-Functional Testing

6.1 Build Verification

Check	Result
<code>npm run build</code>	Compiles successfully, 0 TypeScript errors
Static pages generated	15/15 pages (2 static, 13 dynamic)
Bundle size	Within acceptable range

6.2 Browser Compatibility

Browser	Result
Chrome 126+	PASS — all features functional
Firefox 127+	PASS — all features functional
Safari 18+	PASS — all features functional
Edge 126+	PASS — all features functional

6.3 Responsive Design

Viewport	Result
Desktop (1280x800)	PASS — sidebar navigation, full layouts
Tablet (768x1024)	PASS — sidebar hidden, top nav bar
Mobile (375x812)	PASS — compact nav, stacked cards

6.4 Accessibility

Check	Result
Keyboard navigation	PASS — focus-visible ring on all interactive elements
Form labels	PASS — all inputs have associated labels
Color contrast	PASS — brand colors meet WCAG AA
Screen reader landmarks	PASS — nav, main, aside used correctly

7. Conclusion

All 62 test cases pass after remediation. The application meets functional requirements for the 6 implemented modules (Patient Registration, Appointment Scheduling, Clinical Encounters, Ward Management, Patient Portal, Audit Trail). The 4 modules deferred to Phase 2 (Billing, Pharmacy, Laboratory, Reporting) do not block deployment of the core clinical workflow.

End of QA Test Report